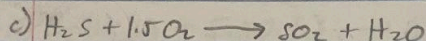
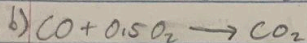


HW #11

- 1) Acetylene has an adiabatic flame temperature of $3,100^{\circ}\text{C}$ while methane has an adiabatic flame temperature of $2,055^{\circ}\text{C}$. Since acetylene has a higher flame temperature, it can melt a larger variety of metals like steel, so even if it's more expensive, less of it would need to be used in order to weld.

22/20

2) Stoichiometric combustion reactions



- For reaction a, since there's 0.40 mol C_2H_2 (40 mol%), this means that required O_2 is $(0.40)(3.5) = 1.40$ mol
- For reaction b, since there's 0.30 mol CO (30 mol%), this means that required O_2 is $(0.30)(0.5) = 0.15$ mol
- For reaction c, since there's 0.01 mol H_2S (1 mol%), this means that required O_2 is $(0.01)(1.5) = 0.015$ mol
- Total O_2 required: $1.40 + 0.15 + 0.015 = 1.565$ mol O_2
- The waste stream already has 0.10 mol O_2 , so the O_2 needed from the air is: $1.565 - 0.10 = 1.465$ mol O_2 ✓

- Air requirement: since air is 21% O_2 and we need 1.465 mol O_2

$$\text{Theoretical air: } \frac{1.465}{0.21} = 6.976$$

Since there's 20% excess air: Air supplied = $(1.2)(6.976) = 8.371$ mol

- So from air supplied, the O_2 supplied is $(8.371)(0.21) = 1.758$ mol and the N_2 supplied is $(8.371)(0.79) = 6.613$ mol

- CO_2 in product stream: $0.40 \cdot \frac{2 \text{ mol CO}_2}{1 \text{ mol C}_2\text{H}_2} = 0.80$ mol CO_2 ✓

$$0.30 \cdot \frac{1 \text{ mol CO}_2}{1 \text{ mol CO}} = 0.30 \text{ mol CO}_2$$

$$0.80 + 0.30 = 1.10 \text{ mol CO}_2$$

- H_2O in product stream: $0.40 \cdot \frac{3 \text{ mol H}_2\text{O}}{1 \text{ mol C}_2\text{H}_2} = 1.20$ mol H_2O

$$0.01 \cdot \frac{1 \text{ mol H}_2\text{O}}{1 \text{ mol H}_2\text{S}} = 0.01 \text{ mol H}_2\text{O}$$

$$1.20 + 0.01 = 1.21 \text{ mol H}_2\text{O}$$

- SO_2 from reaction c also has 0.01 mol in product stream

• Excess O_2 in product stream: $O_{2 \text{ applied}} - O_{2 \text{ needed}} = 1.758 - 1.465 = 0.293 \text{ mol}$

• N_2 is product stream: $6.613 + 0.17 = 6.803$

• Total flue product: $1.10 + 1.21 + 0.01 + 0.293 + 6.803 = 9.416$ ✓

• Heat released in combustion: $Q = \sum n_i (-\Delta H_{c,i}^\circ)$

For C_2H_6 : $-1428.6 \frac{\text{kJ}}{\text{gmol}}$, for CO : $-283 \frac{\text{kJ}}{\text{gmol}}$, for H_2S : $-518.7 \frac{\text{kJ}}{\text{gmol}}$

So $Q_{\text{release}} = (0.40)(1428.6) + (0.30)(-283) + (0.01)(518.7) = 661.53 \text{ kJ/gmol}$

• For heating flue from 25°C to 150°C : $Q_{\text{sens}} = n_i \sum_{298}^{423} C_{p,i}(T) dT$

- For CO_2 : $(1.10)(37.0)(423-298) = 5087.5$

- For H_2O : $(1.21)(33.6)(423-298) = 5082$

- For N_2 : $(6.803)(29.1)(423-298) = 24746$

- For O_2 : $(0.293)(29.3)(423-298) = 1073$

- For SO_2 : $(0.01)(39.9)(423-298) = 50$

$Q = 36038.5 = 36.04 \text{ kJ/gmol}$ ✓

• So the net heat available for steam generation: $Q_{\text{net}} = Q_{\text{release}} - Q_{\text{sens}}$ ✓

So $Q_{\text{net}} = 661.53 - 36.04 = 625.49 \text{ kJ/gmol}$

• Energy required to generate 1 kg steam: $Q_{\text{steam}} = h_g - h_{fw}$

- h_g is enthalpy of saturated steam at 10 bar: $h_g = 2777.1$

- h_{fw} is enthalpy of liquid water at 60°C and 10 bar: h_f at 60°C : 209.32

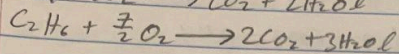
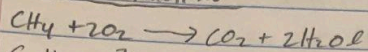
$Q_{\text{steam}} = 2777.1 - 209.32$, so $Q_{\text{steam}} = 2568.77 \text{ kJ/kg}$

• So steam per gmol of waste gas: $\frac{Q_{\text{net}}}{Q_{\text{steam}}}$

So kg steam per gmol of waste gas: $\frac{625.49}{2568.77} = 0.243 \text{ kg steam per gmol}$ ✓

20%

3a) Combustion reactions



O_2 needed for 0.85 mol CH_4 : $(2 \cdot 0.85) = 1.70 \text{ mol O}_2$

O_2 needed for 0.15 mol C_2H_6 : $(\frac{7}{2} \cdot 0.15) = 0.525 \text{ mol O}_2$

So total O_2 : $1.70 + 0.525 = 2.225 \text{ mol O}_2$

CO_2 product from reactions: $0.85 + (2 \cdot 0.15) = 1.15 \text{ mol}$

H_2O products from reactions: $(2 \cdot 0.85) + (3 \cdot 0.15) = 2.15 \text{ mol}$

Heat of combustion using $Q = \sum n_i (-\Delta H_{c,i})$

- For C_2H_6 : $-1428.6 \frac{\text{kJ}}{\text{mol}}$, for CH_4 : $-802.6 \frac{\text{kJ}}{\text{mol}}$

So $Q_{\text{release}} = (0.15)(-1428.6) + (0.85)(-802.6) = -896.5 \frac{\text{kJ}}{\text{mol}}$

Since $\dot{n} = 1 \text{ mol fuel per second}$, $\dot{Q} = \frac{Q_{\text{release}}}{\dot{n}}$

So $\dot{Q} = \frac{-896.5}{1}$, so $\dot{Q} = -896.5 \frac{\text{kJ}}{\text{mol}} = -\Delta H_R$

3b) Since it's a closed system, $Q = \Delta U_{\text{reaction}}$

$\Delta U_{\text{reaction}} = \Delta H - \Delta(n_{\text{gas}})RT$, where $\Delta n_{\text{gas}} = \text{mol gas products} - \text{mol gas reactants}$

For CH_4 , there's 3 mol gas reactant and 1 mol gas product: $n_{\text{gas}} = 1 - 3 = -2$

For 0.85 mol CH_4 , $\Delta n_{\text{CH}_4} = (0.85)(-2) = -1.70$

For C_2H_6 , there's $\frac{7}{2}$ mol reactant and 2 mol gas product: $n_{\text{gas}} = 2 - 4.5 = -2.5$

For 0.15 mol C_2H_6 , $\Delta n_{\text{C}_2\text{H}_6} = (0.15)(-2.5) = -0.375$

So total $\Delta n_{\text{gas}} = -1.70 - 0.375 = -2.075$

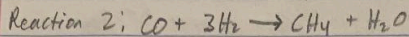
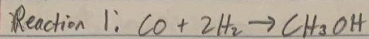
Since H is in kJ , get R to kJ : $R = 0.008314 \frac{\text{kJ}}{\text{mol} \cdot \text{K}}$

So, $\Delta U_{\text{reaction}} = -896.5 - (-2.075)(0.008314)(298) = -891.36 \text{ kJ}$

So $Q = -891.36 \frac{\text{kJ}}{\text{mol fuel}}$

3c) Any excess O_2 goes through the reactor unreacted. It enters and leaves at 25°C so its enthalpy change is zero. All stoichiometric O_2 reacts in the combustion reaction, and is the only O_2 that changes in the standard enthalpy change of reaction. Excess O_2 doesn't affect reaction's enthalpy change.

4) 6.68



75 gmol/s of H_2 and 25 gmol/s CO

So $n_{\text{CO}} = 25 - \xi_1 - \xi_2$

$n_{\text{H}_2} = 75 - 2\xi_1 - 3\xi_2$

$n_{\text{CH}_3\text{OH}} = \xi_1$, $n_{\text{CH}_4} = \xi_2$, $n_{\text{H}_2\text{O}} = \xi_2$

Since 80% of CO fed to reactor is converted into products

$\xi_1 + \xi_2 = 0.80(25)$, so $\xi_1 + \xi_2 = 20$

$n_{\text{CH}_3\text{OH}} = \xi_1$, and if methanol production ($n_{\text{CH}_3\text{OH}}$) is $18.7 \frac{\text{gmol}}{\text{s}}$, then that means $\xi_1 = 18.7$

So to find ξ_2 : $\xi_2 = 20 - \xi_1$, so $\xi_2 = 1.3$ ✓

So $n_{\text{CO}} = 25 - 18.7 - 1.3$, so $n_{\text{CO}} = 5$

$n_{\text{H}_2} = 75 - 2(18.7) - 3(1.3) = 33.7$ ✓

The reactor is adiabatic

$\dot{Q} + \sum n_i \hat{H}_i(T_{\text{in}}) - \sum n_e \hat{H}_e(T_{\text{out}}) = 0$

Since adiabatic, $\dot{Q} = 0$, so $\sum n_i \hat{H}_i(T_{\text{out}}) = \sum n_i \hat{H}_i(T_{\text{in}})$

- Since $\hat{H}_i(T) = \hat{H}_i^\circ(25) + \int_{25}^T c_{p,i} dT$

then: $\sum n_i c_{p,i}(T_{\text{in}}) + \xi_1 \Delta H_{r1}^\circ + \xi_2 \Delta H_{r2}^\circ = 0$, and $\Delta H_r^\circ = \sum \nu_i \Delta H_{f,i}^\circ$

So $\Delta H_{r1}^\circ = H_{f,\text{CH}_3\text{OH}}^\circ - (\Delta H_{f,\text{CO}}^\circ + \Delta H_{f,\text{H}_2}^\circ) = (1)(-200.94) - (1)(-110.53) = -90.41$

$\Delta H_{r2}^\circ = (H_{f,\text{CH}_4}^\circ + \Delta H_{f,\text{H}_2\text{O}}^\circ) - (\Delta H_{f,\text{CO}}^\circ + \Delta H_{f,\text{H}_2}^\circ) = ((1)(-74.52) + (1)(-241.93)) - ((1)(-110.53)) = -205.92 \frac{\text{kJ}}{\text{mol}}$ ✓

So for the heat release: $Q = \xi_1 \Delta H_{r1}^\circ + \xi_2 \Delta H_{r2}^\circ = (18.7)(-90.41) + (1.3)(-205.92)$

So: $Q = -1958.23 \frac{\text{kJ}}{\text{s}}$

Finding $\sum n_i c_{p,i}$: $\text{CO } c_p = 29.1$, $\text{H}_2 c_p = 29.1$, $\text{CH}_3\text{OH } c_p = 43.9$, $\text{CH}_4 c_p = 35.7$

$\text{H}_2\text{O } c_p = 35.6$

So: $5(29.1) + 33.7(29.1) + 18.7(43.9) + 1.3(35.7) + 1.3(35.6) = 2037.19 \frac{\text{kJ}}{\text{K}\cdot\text{s}}$

Getting $\sum n_i c_{p,i}$ into proper units: $2037.2 \frac{\text{kJ}}{\text{K}\cdot\text{s}} = 2.037 \frac{\text{MJ}}{\text{K}\cdot\text{s}}$

$T_{\text{out}} - T_{\text{in}} = \frac{-Q}{\sum n_i c_{p,i}}$, where $T_{\text{in}} = 273^\circ\text{C} = 548\text{K}$

- So $T_{\text{out}} = \frac{-(-1958.23)}{2.037} + 548 = 1509.3\text{K} = \text{reactor at outlet temp}$ ✓

For cold stream in E-1

- the feed going through is 75 mol/s H_2 and 25 mol/s CO

$$\text{So } c_{p,\text{cold}} = 75(29.1) + 25(29.1) = 2910 \text{ J/K.s} = 2.91 \text{ kJ/K.s}$$

$$\Delta T = 275 - 25 = 250 \text{ K}$$

Heat absorbed by cold stream: $Q_{\text{cold}} = c_{p,\text{cold}} \Delta T$

$$Q_{\text{cold}} = (2.91)(250), \text{ so } Q_{\text{cold}} = 727.5 \text{ kJ/s}$$

$$\text{so } Q_{\text{hot}} = -727.5 \text{ kJ/s}$$

H₂ inlet temp: 1509.3 K, and hot mix cp: $c_{p,\text{hot}} = 2.037 \text{ kJ/K.s}$

$$Q_{\text{hot}} = c_{p,\text{hot}} (T_{\text{hot,out}} - T_{\text{hot,in}})$$

$$\text{So: } -727.5 = (2.037)(T_{\text{hot,out}} - 1509.3)$$

$$\text{So } T_{\text{hot,out}} = 1152.2 \text{ K}$$

For the adiabatic reactor equation

$$Q_{\text{reaction}} + Q_{\text{heating of products}} = 0$$

$$Q_{\text{reaction}} = \sum_j \Delta H_{f,j}^\circ$$

$$Q_{\text{heating of products}} = \sum_i n_i c_{p,i} (T_{\text{out}} - T_{\text{in}})$$

$$\text{So } \sum_i n_i c_{p,i} (T_{\text{out}} - T_{\text{in}}) + \sum_j \Delta H_{f,j}^\circ = 0$$

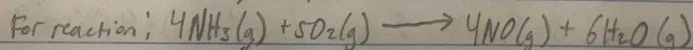
$$5) \text{ Heat capacity } c_p(T) = A + BT + CT^2 + DT^3 + \frac{E}{T^2}$$

$$\text{So enthalpy relative to 298 K: } H(T) - H(298) = A(T - 298) + \frac{B}{2}(T^2 - 298^2) + \frac{C}{3}(T^3 - 298^3) + \frac{D}{4}(T^4 - 298^4) - E\left(\frac{1}{T} - \frac{1}{298}\right)$$

$$\text{Using excel, } H_{NH_3}(473) - H_{NH_3}(298) = 6.70 \text{ kJ/mol}$$

$$\text{Using excel, } H_{O_2}(473) - H_{O_2}(298) = 5.25 \text{ kJ/mol}$$

$$\Delta H_{\text{reaction}}(298) = \sum_{\text{products}} H_f^\circ(\text{products}) - \sum_{\text{reactants}} H_f^\circ(\text{reactants})$$



$$\text{So } \Delta H_{\text{reaction}}(298) = (4(90.25) + 6(-241.83)) - (4(-46.15)) = -905.38 \text{ kJ}$$

$$H_{\text{reactants}}: (4)(6.70) + (5)(5.25) = 58.3 \text{ kJ/s}$$

6 mol/s O_2 is fed, 5 are used stoichiometrically, so 1 mol/s O_2 is product stream

$$H_{\text{products}}(T) = 4h_{NO}(T) + 6h_{H_2O}(T) + 1h_{O_2}(T)$$

$$H_{\text{products}}(T) - H_{\text{reactants}} = -\Delta H_{\text{reaction}}$$

$$\text{So } H_{\text{products}}(T) - 58.3 = -905.38, \text{ so } H_{\text{products}}(T) = 963.38$$

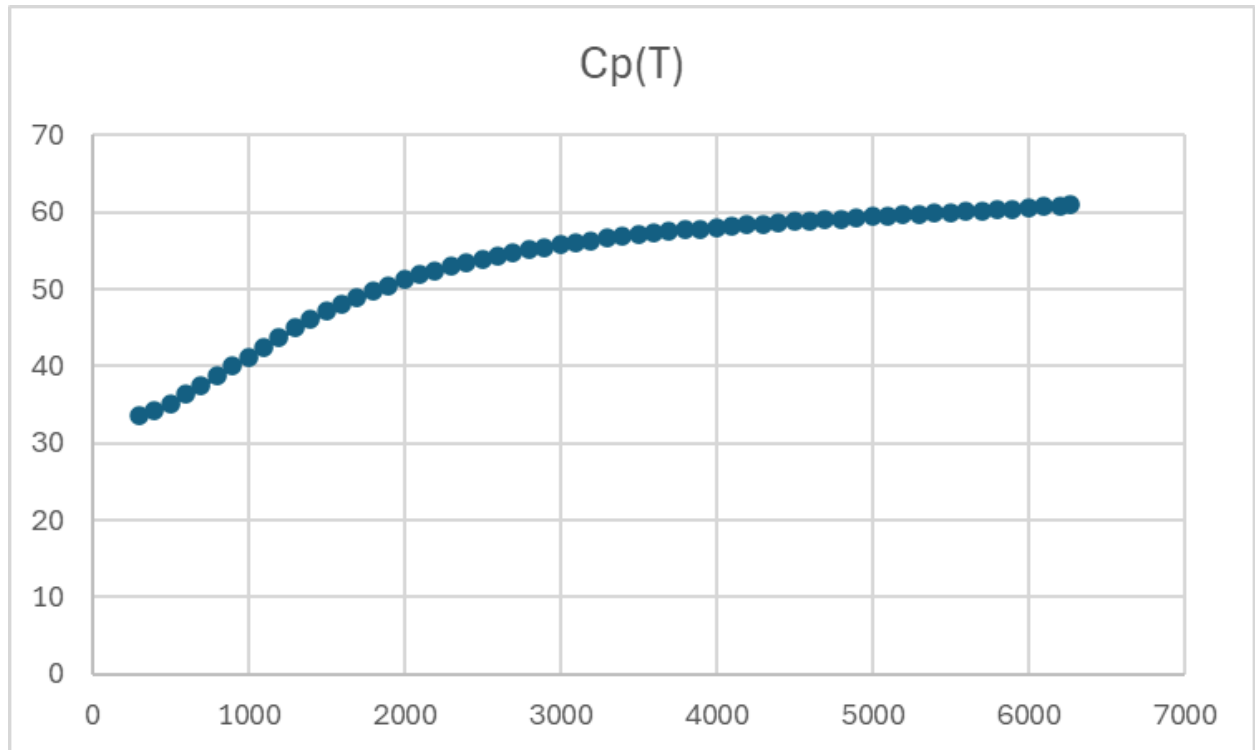
Using coefficients for higher temps, $T = 2470.85 \text{ K}$

My Showmate result was $c_p = 33.589 \frac{\text{J}}{\text{mol} \cdot \text{K}}$ at 298K

At 298K, c_p of water vapor is $33.6 \frac{\text{J}}{\text{mol} \cdot \text{K}}$

Showmate equation gave good results at 298K

Only one solution is found because the $c_p(T)$ polynomial becomes inaccurate above 2000K. Going beyond that makes incorrect enthalpy shapes, so there's one mathematical but one physical solution



There was no other graph to compare it to. I'm not sure what the question was asking.